Black Box System Testing of the Material Planning System

Experiment 5 | English technical report | 22 October 2026

This report evaluates the implemented material planning system against its user requirements through external HTTP requests and real browser operations. The final HTTP run completed 125 cases, with 125 passing. Browser suites supplied 65 additional passing checks. A CSV field-length defect found by boundary testing was corrected and retested. An intermittent Windows connection reset on an oversized request remains a documented transport observation.

Table 1.1 Report identity

Item	Value
Project	Design and Implementation of a Production Requirements and Procurement Planning System Based on Multilevel Bills of Materials
Method	Structured procedural implementation; equivalence partitioning and boundary value analysis
Group	2 members; [Group number], [Member A name and student ID], [Member B name and student ID]
Test scope	Source maintenance S1; integrated planning S2 to S5; history and export S6
Status	Technical execution and reporting complete within stated scope; actual group review and contributions pending

1 Objectives and requirements

The E5 guide requires at least as many important modules or subsystems as group members, explicit preconditions and postconditions, test cases combining equivalence partitioning and boundary value analysis, and an English report. The three selected functional groups cover the six existing subsystems and exceed the two-member minimum. Grouping S2 to S5 into one external calculation scenario does not redefine the E2 architecture.

Expected results come from the written BR01 to BR09 rules, FR01 to FR08 acceptance conditions, CSV limits and persistence behavior. Internal coverage, private functions and database tables are not used to decide pass or fail. Source files were inspected only to identify public interfaces and diagnose observed defects; the black-box runner imports no application modules.

2 Selected subsystems and contracts

Table 2.1 Preconditions and postconditions

Group	Preconditions	Postconditions
S1 Source maintenance	Current source/request records; named UTF-8 CSV tables or JSON source save; expected previous revision for saving.	Preview either returns READY with a complete candidate or INVALID with located errors. Preview never saves. Valid save creates a version; invalid or stale save preserves current data.
S2 to S5 Integrated planning	Selected task IDs; complete material, BOM and stock records; valid quantities, dates and rule version; matching stock/baseline date.	VALID returns net production, dated purchasing, stock/source trace and risk; a complete batch is saved. INVALID returns blocking errors without a saved batch. Persisted inventory is not deducted.
S6 History and export	Existing saved batch ID, supported export kind or a nonnegative history offset. Source records may have changed since the batch was created.	History returns the original snapshot and result. CSV uses that batch and carries its ID. Missing batches and invalid requests reject. Restart retains saved state.

Invalid inputs intentionally violate these preconditions one at a time where practical. Their expected postcondition is an informative rejection with no partial persistence. Cross-record consistency cases cover relationships that individual field partitions cannot express.

2.1 Acceptance limits

Task demand and BOM usage are positive integers; stock and lead time are nonnegative integers; priority is 1, 2 or 3. Reserved stock must not exceed on-hand stock. CSV numeric inputs and the browser editor are restricted to the JavaScript safe-integer maximum 9007199254740991. The native JSON calculation interface uses Python integers and does not impose that same upper bound. These are separate entry-point contracts.

Each CSV file is limited to 500000 UTF-8 bytes and 5000 records. The combined HTTP JSON request is limited to 2000000 bytes. Boundary tests keep other constraints valid: the row-limit fixture stays below the byte limit, while the file-size fixture has only two material rows and uses one long name field.

3 Test environment and independent fixtures

Execution: 2026-10-22T12:29:30.179266+08:00. Python 3.12.14 (main, Aug 25 2026, 14:01:42) [MSC v.1944 64 bit (AMD64)]. Platform Windows-11-10.0.26200-SP0. The runner starts the real

loopback HTTP server on an unused local port, creates a new test database under results/e5-runs, and stops only that server. It does not open or write the production planning.sqlite3 database.

All 371 recorded HTTP exchanges have request/response artifacts, except a transport exception would instead have an explicit error record. Request bodies preserve exact JSON or CSV bytes. Test IDs bind the readable case register to those artifacts. The final report is based on a completed run, not a partial result emitted during cleanup.

Table 3.1 Base fixture

Record	Fields
Material P	MAKE product; unit pcs; lead time 2 days
Material R	BUY raw material; unit pcs; lead time 3 days; minimum 0; purchase multiple 5
BOM	P requires 2 units of R
Stocks	P and R each have on_hand 0 and reserved 0; snapshot date 2026-11-01
Task T1	Product P; quantity 10; due 2026-11-20; priority 2; selected
Request	Baseline 2026-11-01; rule version prototype-v0.2; request ID E5

Independent base oracle: P has no stock, so manufacture 10. Start P on 18 November. Raw demand is 10 times 2, or 20. No raw stock exists; 20 already meets the multiple of 5, so purchase 20 for arrival on 18 November and order on 15 November. No date precedes the baseline.

The BUY-root variant removes P and all BOM edges, keeps R and its stock, and points T1 directly to R. The multilevel fixture adds S: P requires 2 S and S requires 3 R. With P stock 2, S stock 4 and R on_hand 5/reserved 1, demand10 gives P8, S12, raw gross36, raw gap32 and purchase35. S lead time1 makes the raw arrival17 November and order14 November.

4 Equivalence partitions and boundary selections

Table 4.1 Partition model

Partition	Valid classes	Invalid classes	Witness families
EP01 Quantities	Positive native JSON integers; nonnegative stock; CSV/browser safe integers	Zero demand or usage; negative stock; decimal, Boolean or text quantity; CSV number above safe maximum	Q, B, ST, RS, NUM
EP02 Task order	Priority1 to3; known selected tasks; priority/date/ID tie-breaking	Priority0 or4; empty/unknown selection	PR, ORD, INV00 to01

Partition	Valid classes	Invalid classes	Witness families
EP03 BOM graph	Known acyclic references; MAKE parents and BUY leaves	Unknown/duplicate edge; self/multiple-node cycle; missing BOM/stock; BUY with children	INV02 to08, NET
EP04 Purchasing	Positive gaps; exact/rounded lots; zero gap; same/later-date surplus	Nonpositive multiple; negative minimum; future surplus used backward is forbidden	MU, LOT, SUR
EP05 Dates	Canonical calendar dates; leap day; dates before/on/after baseline	Impossible/noncanonical date; backward arithmetic below supported date range	LD, DT, CAL, INV09
EP06 CSV input	Four exact headers; valid quoted UTF-8 text; supported table set	Empty selection/file; unknown table; bad headers/columns/quotes/numbers; row/byte excess	CSV, ROW, BYTE, NUM
EP07 Persistence	New/current revision; existing batch; supported export; nonnegative offset	Stale revision; invalid save; absent batch; invalid kind or offset	H, EX, HX, PAGE, RESTART

Table 4.2 Boundary value matrix

Boundary	Values executed	Reason
Demand lower bound	0, 1, 2	Just below, at and above positive lower bound
CSV safe-integer maximum	9007199254740990, 9007199254740991, 9007199254740992	Below, at and above entry-point maximum
Priority	0, 1, 2, 3, 4	Outside both ends and all valid discrete classes
Stock versus demand20	0, 19, 20, 21; negative stock separately	Shortage, exact coverage and surplus
Reserved with on-hand20	-1, 0, 1, 20, 21	Lower and cross-field upper limits
Lot multiple10	Gap9, 10, 11; minimum9, 10, 11 with gap1	Below, exact and above a lot threshold
Order versus baseline	One day before, same day, one day after	Risk is strict less-than, not less-than-or-equal
CSV records	4999, 5000, 5001	Record count maximum
CSV UTF-8 bytes	499999, 500000, 500001	Exact file byte maximum
HTTP body bytes	1999999, 2000000, 2000001; empty separately	Combined request maximum
History offsets	0, 99, 100, 101, 102 with 101 batches	100-row page boundary and empty tail

EP denotes equivalence partitioning; BVA denotes boundary value analysis. Combined labels describe a case that is both a representative of a partition and a boundary witness. They do not imply every

boundary of every input field has been tested. Invalid cases use an otherwise valid base to avoid masking one defect with another.

5 Requirement to test traceability

Table 5.1 Functional coverage mapping

Req ID	Acceptance focus	Test IDs
FR01	Source maintenance and import	S00, Q, PR, RS, LD, MU, CAL, INV09 to11, CSV, ROW, BYTE, NUM, H00 to03
FR02	Task selection and deterministic order	PR, INV00 to01, ORD0 to4
FR03	BOM and snapshot validity	B, INV02 to08
FR04	Multilevel net requirements	P00, Q, B, NET0 to1, EX5
FR05	Shared stock and no actual deduction	ST, RS, NET, ORD, H08
FR06	Purchasing and dated planned surplus	P00, Q, ST, MU, LOT, NET1, SUR, H06
FR07	Calendar dates and timing risks	P00, LD, DT, CAL, NET1
FR08	History, trace and export	H00 to08, HX, EX, PAGE, RESTART and browser suites

NFR02 reproducibility is checked by ORD3, which reverses input rows. NFR07 data integrity is checked by rejection-side history invariants, H01, H03, H08 and RESTART. Performance was measured in an earlier phase and is not claimed as a new E5 measurement. This map is requirements coverage, not source-code coverage.

6 Executed HTTP case register

Each row reports a specific executed case. Base fixture fields remain unchanged unless the scenario states otherwise. The evidence files contain exact inputs and complete outputs. VALID cases additionally require HTTP200 and a saved batch. INVALID planning cases additionally require HTTP200, at least one located error, unchanged history, and no saved_batch field. CSV cases additionally require HTTP200 and unchanged saved workspace and history. These repeated checks are abbreviated in the tables.

Output notation uses JSON arrays. Purchase rows are [material, arrival date, quantity, order date]; make rows are [material, quantity, start date]. Empty arrays mean no rows. Boolean true means the

named invariant was observed. Dynamic revision IDs are shown in full in the evidence; readable tables use the word revision where appropriate.

6.1 Planning cases

Table 6.1 Planning expected and observed results

ID	Scenario and input	Expected	Observed
P00	Base P10 with two R per P: make10, buy20; calendar lead2+3	VALID + saved; [[["P", 10, "2026-11-18"]], [["R", "2026-11-18", 20, "2026-11-15"]], []]	VALID + saved; [[["P", 10, "2026-11-18"]], [["R", "2026-11-18", 20, "2026-11-15"]], []] | PASS
Q00	JSON task quantity 0; native positive integer domain; UI safe limit is separate	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
Q01	JSON task quantity 1; native positive integer domain; UI safe limit is separate	VALID + saved; [1, 5]	VALID + saved; [1, 5] | PASS
Q02	JSON task quantity 2; native positive integer domain; UI safe limit is separate	VALID + saved; [2, 5]	VALID + saved; [2, 5] | PASS
Q03	JSON task quantity 9007199254740990; native positive integer domain; UI safe limit is separate	VALID + saved; [9007199254740990, 18014398509481980]	VALID + saved; [9007199254740990, 18014398509481980] | PASS
Q04	JSON task quantity 9007199254740991; native positive integer domain; UI safe limit is separate	VALID + saved; [9007199254740991, 18014398509481985]	VALID + saved; [9007199254740991, 18014398509481985] | PASS
Q05	JSON task quantity 9007199254740992; native positive integer domain; UI safe limit is separate	VALID + saved; [9007199254740992, 18014398509481985]	VALID + saved; [9007199254740992, 18014398509481985] | PASS
Q06	JSON task quantity 1.5; native positive integer domain; UI safe limit is separate	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
Q07	JSON task quantity True; native positive integer domain; UI safe limit is separate	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
Q08	JSON task quantity '10'; native positive integer domain; UI safe limit is separate	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
B00	BOM usage 0; positive lower boundary	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS

ID	Scenario and input	Expected	Observed
B01	BOM usage 1; positive lower boundary	VALID + saved; 10	VALID + saved; 10 | PASS
B02	BOM usage 2; positive lower boundary	VALID + saved; 20	VALID + saved; 20 | PASS
PR0	Priority 0; closed domain 1 to 3	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
PR1	Priority 1; closed domain 1 to 3	VALID + saved; ["T1"]	VALID + saved; ["T1"] | PASS
PR2	Priority 2; closed domain 1 to 3	VALID + saved; ["T1"]	VALID + saved; ["T1"] | PASS
PR3	Priority 3; closed domain 1 to 3	VALID + saved; ["T1"]	VALID + saved; ["T1"] | PASS
PR4	Priority 4; closed domain 1 to 3	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
ST0	Raw stock -1 around gross requirement20	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
ST1	Raw stock 0 around gross requirement20	VALID + saved; 20	VALID + saved; 20 | PASS
ST2	Raw stock 19 around gross requirement20	VALID + saved; 5	VALID + saved; 5 | PASS
ST3	Raw stock 20 around gross requirement20	VALID + saved; 0	VALID + saved; 0 | PASS
ST4	Raw stock 21 around gross requirement20	VALID + saved; 0	VALID + saved; 0 | PASS
RS0	Reserved -1 with on-hand20; allowed interval0..20	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
RS1	Reserved 0 with on-hand20; allowed interval0..20	VALID + saved; 0	VALID + saved; 0 | PASS
RS2	Reserved 1 with on-hand20; allowed interval0..20	VALID + saved; 5	VALID + saved; 5 | PASS
RS3	Reserved 20 with on-hand20; allowed interval0..20	VALID + saved; 20	VALID + saved; 20 | PASS
RS4	Reserved 21 with on-hand20; allowed interval0..20	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS

ID	Scenario and input	Expected	Observed
LD0	Purchase lead time -1; nonnegative lower boundary	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
LD1	Purchase lead time 0; nonnegative lower boundary	VALID + saved; "2026-11-18"	VALID + saved; "2026-11-18" | PASS
LD2	Purchase lead time 1; nonnegative lower boundary	VALID + saved; "2026-11-17"	VALID + saved; "2026-11-17" | PASS
MU0	Purchase multiple 0; positive lower boundary	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
MU1	Purchase multiple 1; positive lower boundary	VALID + saved; 20	VALID + saved; 20 | PASS
MU2	Purchase multiple 2; positive lower boundary	VALID + saved; 20	VALID + saved; 20 | PASS
LOT0	BUY root gap9, minimum0, multiple10	VALID + saved; 10	VALID + saved; 10 | PASS
LOT1	BUY root gap10, minimum0, multiple10	VALID + saved; 10	VALID + saved; 10 | PASS
LOT2	BUY root gap11, minimum0, multiple10	VALID + saved; 20	VALID + saved; 20 | PASS
LOT3	BUY root gap1, minimum9, multiple10	VALID + saved; 10	VALID + saved; 10 | PASS
LOT4	BUY root gap1, minimum10, multiple10	VALID + saved; 10	VALID + saved; 10 | PASS
LOT5	BUY root gap1, minimum11, multiple10	VALID + saved; 20	VALID + saved; 20 | PASS
LOT6	BUY root gap1, minimum-1, multiple10	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
LOT7	Zero shortage with minimum100 creates no purchase	VALID + saved; []	VALID + saved; [] | PASS
DT0	Order date one day before/on/after baseline; due=2026-11-05	VALID + saved; [["2026-10-31", 1]]	VALID + saved; [["2026-10-31", 1]] | PASS
DT1	Order date one day before/on/after baseline; due=2026-11-06	VALID + saved; []	VALID + saved; [] | PASS
DT2	Order date one day before/on/after baseline; due=2026-11-07	VALID + saved; []	VALID + saved; [] | PASS
CAL0	Calendar partition 2026-02-30; leap, format and representable range	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS

ID	Scenario and input	Expected	Observed
CAL1	Calendar partition 2028-02-29; leap, format and representable range	VALID + saved; "2028-02-24"	VALID + saved; "2028-02-24" | PASS
CAL2	Calendar partition 20261120; leap, format and representable range	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
CAL3	Calendar partition 0001-01-01; leap, format and representable range	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
CAL4	Calendar partition 9999-12-31; leap, format and representable range	VALID + saved; "9999-12-26"	VALID + saved; "9999-12-26" | PASS
INV00	Empty selected task set	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV01	Unknown selected task	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV02	Unknown BOM reference	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV03	Duplicate BOM edge	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV04	Self cycle	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV05	Two-node cycle even when finished stock covers task	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV06	MAKE material without children	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV07	BUY material with children	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV08	Reachable material missing stock	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV09	Snapshot date differs from baseline	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS

ID	Scenario and input	Expected	Observed
INV10	Duplicate material ID	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
INV11	Blank required identifier	INVALID; located errors; no saved plan or history change	INVALID; located errors; no saved plan or history change | PASS
NET0	Finished stock exactly covers demand; no lower requirements	VALID + saved; [[], [], 1]	VALID + saved; [[], [], 1] | PASS
NET1	Multilevel net explosion: P8, S12, raw gap32 rounded35	VALID + saved; [[["P", 8, "2026-11-18"], ["S", 12, "2026-11-17"]], [["R", "2026-11-17", 35, "2026-11-14"]]]	VALID + saved; [[["P", 8, "2026-11-18"], ["S", 12, "2026-11-17"]], [["R", "2026-11-17", 35, "2026-11-14"]]] | PASS
ORD0	Shared stock ordering by priority	VALID + saved; [["A", "Z"], [["A", 4], ["Z", 1]]]	VALID + saved; [["A", "Z"], [["A", 4], ["Z", 1]]] | PASS
ORD3	Reversed input rows preserve business allocation	VALID + saved; [["A", "Z"], [["A", 4], ["Z", 1]]]	VALID + saved; [["A", "Z"], [["A", 4], ["Z", 1]]] | PASS
ORD4	Unselected high-priority task consumes no stock	VALID + saved; [["Z"], [["Z", 4]]]	VALID + saved; [["Z"], [["Z", 4]]] | PASS
ORD1	Shared stock ordering by date	VALID + saved; [["Z", "A"], [["Z", 4], ["A", 1]]]	VALID + saved; [["Z", "A"], [["Z", 4], ["A", 1]]] | PASS
ORD2	Shared stock ordering by identifier	VALID + saved; [["A", "Z"], [["A", 4], ["Z", 1]]]	VALID + saved; [["A", "Z"], [["A", 4], ["Z", 1]]] | PASS
SUR0	Earlier purchase surplus covers later demand despite priority order	VALID + saved; [[["R", "2026-11-10", 10, "2026-11-07"], ["R", "2026-11-11", 0, null]], [4]]	VALID + saved; [[["R", "2026-11-10", 10, "2026-11-07"], ["R", "2026-11-11", 0, null]], [4]] | PASS
SUR1	Future lot cannot supply earlier demand; chronology reversed	VALID + saved; [["R", "2026-11-09", 10, "2026-11-06"], ["R", "2026-11-10", 0, null]]	VALID + saved; [["R", "2026-11-09", 10, "2026-11-06"], ["R", "2026-11-10", 0, null]] | PASS
SUR2	Same-date shortages merge before rounding	VALID + saved; [["R", "2026-11-09", 10, "2026-11-06"]]	VALID + saved; [["R", "2026-11-09", 10, "2026-11-06"]] | PASS
H04	Save baseline plan for historical export	VALID + saved; 20	VALID + saved; 20 | PASS
H06	New plan reflects changed demand11	VALID + saved; 25	VALID + saved; 25 | PASS

6.2 Source cases

Table 6.2 Source expected and observed results

ID	Scenario and input	Expected	Observed
S00	Fresh workspace has no saved revision	No saved workspace	No saved workspace | PASS
CSV00	Valid four tables; preview does not save or create a plan	"READY"; workspace and history unchanged	"READY"; workspace and history unchanged | PASS
CSV01	Empty file selection	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV02	Empty tasks file	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV03	Unknown table	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV04	Missing header	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV05	Duplicate header	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV06	Too few columns	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV07	Unclosed quote	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV08	Decimal quantity	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV09	Scientific quantity	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV10	Quoted Chinese comma quote and newline preserved	["READY", "产品, \"甲\"\n第二行"]; workspace and history unchanged	["READY", "产品, \"甲\"\n第二行"]; workspace and history unchanged | PASS
NUM0	CSV quantity 9007199254740990; safe-integer maximum boundary	"READY"; workspace and history unchanged	"READY"; workspace and history unchanged | PASS
NUM1	CSV quantity 9007199254740991; safe-integer maximum boundary	"READY"; workspace and history unchanged	"READY"; workspace and history unchanged | PASS
NUM2	CSV quantity 9007199254740992; safe-integer maximum boundary	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
ROW0	CSV row count 4999 around5000 maximum	"READY"; workspace and history unchanged	"READY"; workspace and history unchanged | PASS
ROW1	CSV row count 5000 around5000 maximum	"READY"; workspace and history unchanged	"READY"; workspace and history unchanged | PASS

ID	Scenario and input	Expected	Observed
ROW2	CSV row count 5001 around5000 maximum	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
BYTE0	CSV bytes 499999 around500000 maximum	"READY"; workspace and history unchanged	"READY"; workspace and history unchanged | PASS
BYTE1	CSV bytes 500000 around500000 maximum	"READY"; workspace and history unchanged	"READY"; workspace and history unchanged | PASS
BYTE2	CSV bytes 500001 around500000 maximum	"INVALID"; workspace and history unchanged	"INVALID"; workspace and history unchanged | PASS
CSV11	Template ZIP contains four data tables	[200, ["bom.csv", "materials.csv", "stocks.csv", "tasks.csv"]]	[200, ["bom.csv", "materials.csv", "stocks.csv", "tasks.csv"]] | PASS

6.3 History cases

Table 6.3 History expected and observed results

ID	Scenario and input	Expected	Observed
H00	First save from null revision creates a source version	[200, "SAVED", true]	[200, "SAVED", true] | PASS
H01	Invalid save preserves last valid revision	200; INVALID; original revision unchanged	200; INVALID; original revision unchanged | PASS
H02	Matching revision permits new version	[200, "SAVED", true]	[200, "SAVED", true] | PASS
H03	Stale revision fails without overwriting newer source	[409, "CONFLICT", 11]	[409, "CONFLICT", 11] | PASS
EX0	MAKE download rows and batch ID; zero-risk header-only partition	1; workspace and history unchanged	1; workspace and history unchanged | PASS
EX1	PURCHASE download rows and batch ID; zero-risk header-only partition	1; workspace and history unchanged	1; workspace and history unchanged | PASS
EX2	ALLOCATION download rows and batch ID; zero-risk header-only partition	2; workspace and history unchanged	2; workspace and history unchanged | PASS
EX3	TRACE download rows and batch ID; zero-risk header-only partition	2; workspace and history unchanged	2; workspace and history unchanged | PASS
EX4	RISK download rows and batch ID; zero-risk header-only partition	0; workspace and history unchanged	0; workspace and history unchanged | PASS
EX5	Export agrees with independent make10 and start date	[[10, "2026-11-18"]]	[[10, "2026-11-18"]] | PASS
H07	Old complete snapshot and CSV unchanged after new plan	[true, true]	[true, true] | PASS

ID	Scenario and input	Expected	Observed
H08	Calculating plans does not deduct persisted stock	[[0, 0], [0, 0]]	[[0, 0], [0, 0]] | PASS
HX0	/api/history/missing	404	404 | PASS
HX1	/api/history/missing/csv/MAKE	404	404 | PASS
HX2	/api/history/4c25c33172a8470e97d2d50c9221eaa7/csv/OTHER	400	400 | PASS
HX3	/api/history?offset=-1	400	400 | PASS
HX4	/api/history?offset=abc	400	400 | PASS
PAGE0	History offset0 with 101 saved plans	[200, 100, 100]	[200, 100, 100] | PASS
PAGE1	History offset99 with 101 saved plans	[200, 2, null]	[200, 2, null] | PASS
PAGE2	History offset100 with 101 saved plans	[200, 1, null]	[200, 1, null] | PASS
PAGE3	History offset101 with 101 saved plans	[200, 0, null]	[200, 0, null] | PASS
PAGE4	History offset102 with 101 saved plans	[200, 0, null]	[200, 0, null] | PASS
RESTART	Restart retains workspace and all saved history	[true, true]	[true, true] | PASS

6.4 Transport cases

Table 6.4 Transport expected and observed results

ID	Scenario and input	Expected	Observed
HTTP0	Empty request body	413	413 | PASS
HTTP1	Malformed JSON	400	400 | PASS
HTTP2	Wrong content type	415	415 | PASS
HTTP3	HTTP UTF-8 body 1999999 bytes; status only above limit	[200, "INVALID"]	[200, "INVALID"] | PASS
HTTP4	HTTP UTF-8 body 2000000 bytes; status only above limit	[200, "INVALID"]	[200, "INVALID"] | PASS
HTTP5	HTTP UTF-8 body 2000001 bytes; status only above limit	[413, null]	[413, null] | PASS

7 Browser workflows and regression evidence

The API cases exercise the complete running server. Three existing browser suites were also rerun with headless Microsoft Edge and isolated temporary databases. They use actual controls, file upload/download, page reload and server restart. These checks are supplementary scenarios, not a claim of exhaustive browser compatibility.

Table 7.1 Browser suite results

Suite	Passed	Recorded time	Principal workflow
Workbench	30/30	2026-10-22T04:26:36.482Z	Edit source, choose tasks, calculate, reject bad input, preserve large displayed integers and check narrow-screen layout
Persistence	14/14	2026-10-22T04:26:41.137Z	Save, reload, view immutable history, handle stale writer and restart the service
CSV browser	21/21	2026-10-22T04:28:19.331Z	Download templates, upload, preview, apply, save, calculate and download all five result types

All 30 workbench, 14 persistence and 21 CSV browser checks passed. The CSV browser suite was rerun after the field-size fix. The 38 existing CSV parsing/API regression checks also passed after that fix. Counts from these suites overlap in business scenarios and must not be summed into a claim about unique requirements.

Table 7.2 Workbench check register

No	Observed scenario	Result
1	Initial material rows	PASS
2	Priority order	PASS
3	Higher priority B net production	PASS
4	Lower priority B net production	PASS
5	Edit invalidates old results	PASS
6	Old result tables removed	PASS
7	Edited LOW quantity recalculated	PASS
8	Decimal input rejected	PASS
9	Error location focuses field	PASS
10	Unsafe integer input rejected	PASS
11	Large calculated quantity displayed exactly	PASS
12	Material row addition	PASS
13	Material row deletion	PASS

No	Observed scenario	Result
14	BOM row addition	PASS
15	BOM row deletion	PASS
16	Stock row addition	PASS
17	Stock row deletion	PASS
18	Production task row addition	PASS
19	Production task row deletion	PASS
20	Empty task selection rejected	PASS
21	Invalid reserved stock surfaced	PASS
22	Two purchase date buckets	PASS
23	Later bucket buys zero	PASS
24	Early-date risk visible	PASS
25	Stale response rejected	PASS
26	Network failure allows retry	PASS
27	Retry succeeds	PASS
28	Mobile no page overflow	PASS
29	Mobile calculate reachable	PASS
30	No uncaught browser errors	PASS

Table 7.3 Persistence check register

No	Observed scenario	Result
1	First save creates source version	PASS
2	Reload restores saved edit	PASS
3	Plan saved badge shown	PASS
4	History shows old LOW quantity	PASS
5	History does not replace current edit	PASS
6	History includes original source snapshot	PASS
7	Conflict keeps unsaved local edit	PASS
8	Conflict does not overwrite database	PASS
9	Invalid save keeps previous data	PASS
10	Explicit reload restores valid data	PASS
11	Server restart restores source	PASS
12	Server restart preserves history	PASS

No	Observed scenario	Result
13	Persistence toolbar and history fit mobile	PASS
14	No unhandled page errors	PASS

Table 7.4 CSV browser check register

No	Observed scenario	Result
1	Template ZIP downloads	PASS
2	ZIP signature	PASS
3	Preview does not replace editor	PASS
4	Preview does not save database	PASS
5	Apply installs imported row	PASS
6	Apply leaves save explicit	PASS
7	Download identifies saved batch	PASS
8	CSV contains independent expected production	PASS
9	PURCHASE downloads UTF8 BOM	PASS
10	ALLOCATION downloads UTF8 BOM	PASS
11	TRACE downloads UTF8 BOM	PASS
12	RISK downloads UTF8 BOM	PASS
13	Edit clears stale export control	PASS
14	Historical download unaffected by new input	PASS
15	Bad CSV keeps original edit	PASS
16	Bad CSV cannot apply	PASS
17	Bad header error visible	PASS
18	Editor change invalidates preview	PASS
19	Invalid encoding cannot apply	PASS
20	Import toolbar fits narrow screen	PASS
21	No unhandled browser errors	PASS

Browser evidence includes results/workbench-results.png, results/persistence-history.png, results/csv-workbench.png and the corresponding mobile screenshots. These images were captured from test databases; they are evidence of those scenarios, not user production records.

8 Findings corrections and residual observations

Table 8.1 Findings and disposition

ID	Finding	Disposition
DEF01	A valid 499999-byte or 500000-byte CSV with one long name field returned CSV_SYNTAX. The underlying csv reader default field limit was 131072 characters.	Fixed csv_io.py to align the field parser limit with the existing 500000-byte file ceiling. BYTE0 and BYTE1 now pass; BYTE2 still rejects. Existing CSV and browser regressions pass.
OBS01	The initial Q05 oracle applied the browser/CSV safe-integer ceiling to the native JSON calculation API. That API accepted the larger integer and returned exact calculated values.	Corrected the oracle using the entry-point contract. Q05 tests exact native JSON arithmetic; NUM0 to2 separately verify the CSV safe boundary. Browser regression checks its own limit.
OBS02	An oversized 2000001-byte request intermittently caused Windows connection abortion while the server was returning413. It interrupted two earlier runner attempts.	Final full run observed413 and passed the status-only boundary criterion. A reliable complete error-body delivery is not established. The harness records body-read errors or a failed transport result rather than silently omitting the case. This compatibility observation is not marked fixed.

The first attempt recorded 118 passing cases out of 121 recorded before interruption; three mismatches included the two CSV failures and the Q05 oracle issue. The preserved blackbox-e5-initial.json and that run directory retain this evidence. The final result contains completed=true and all125 cases. A partial cleanup report must never be mistaken for a completed test run.

The correction changes only CSV parser configuration and its shared byte-limit constant. No calculation rule, stock ordering or user database was changed. E3 and earlier reports are dated design snapshots; final report integration should refresh source line references and hashes affected by the CSV edit.

9 Result interpretation and limitations

The selected cases support the stated quantity, stock, BOM, date, import, persistence and export requirements. Expected values are concrete outputs derived before execution from the business rules. Where full snapshots are compared, a mismatch is visible even if headline quantities are unchanged. Invalid input tests also verify that rejection does not create a history batch.

Passing these cases does not prove correctness for every graph or all inputs. This phase does not exhaust concurrent HTTP clients, disk-full failures, every CSV encoding, every graph depth, alternative browsers or operating systems. Version-conflict behavior is tested with stale revisions and two browser pages, not a high-load concurrency benchmark. The all-member academic review and personal contribution statements remain pending.

Transport OBS02 is supplementary to the selected business cases but remains visible in the report. The final status-only pass does not erase an earlier intermittent failure or establish that every client receives the explanatory error body. A production deployment would require a separate server and compatibility acceptance scope; this project remains a local coursework workbench.

10 Reproducibility and evidence index

Table 10.1 Evidence locations relative to the project root

Location	Content
material-planning/verify_blackbox.py	External HTTP runner and independent fixtures; no application imports
material-planning/results/blackbox-e5.json	Final completed case register, actual/expected results, source and harness hashes
material-planning/results/e5-runs/59ed88f176ed468792caaeea2e4256d1	Exact request and response bytes, exchanges.json, test database and server log
specs/material-requirements-planning/e5/evidence	Frozen copies of the initial/final HTTP results and browser/CSV regression results
specs/material-requirements-planning/e5/verification-report.json	Document/evidence/source consistency checks and final artifact hash
specs/material-requirements-planning/e5/qa	Internal Word-exported PDF and page images used for visual review

Run verify_blackbox.py with the recorded bundled Python runtime; it selects an unused local port and creates a fresh test directory. Run verify_persistence_ui.cjs and verify_csv_ui.cjs with the recorded Node runtime and bundled Playwright/Edge environment; the persistence script also runs the workbench suite. Then run verify_csv.py, rebuild this report, and repeat document verification and page inspection. The scripts stop only the servers they created.

The test run is reproducible in behavior, but batch IDs, revision IDs, port, timestamps and run directory names are intentionally fresh. Equality or invariance checks compare values within the same run. The saved artifacts allow exact reconstruction of a previously executed request.

CSV source SHA256: 4cf5f8c0899bb0cc40e2111c8006a5fba7f926727ed3b212e1ac82e71d51e23e

HTTP runner SHA256: f0315100f587d1347275cf94f57dde37b8ab365bfed1962ff1402ee8acb15de8

11 Group records to complete

The group has two members. The following records must describe work actually performed; automated technical execution is not evidence of who authored, reviewed or approved the assignment.

Table 11.1 Actual task allocation

Member	Identity	Actual task and evidence
A	[Name and student ID]	[Actual requirements analysis, partition design, case execution or report work; cite IDs and dates]
B	[Name and student ID]	[Actual boundary design, independent checking, browser testing or report work; cite IDs and dates]

11.1 Literal review meeting record

Meeting date and place: [To be completed after the meeting]. Participants: [Both actual members]. Chair and recorder: [Actual names].

Review agenda: confirm selected subsystem contracts; check independent fixture calculations; verify partition and boundary choices; inspect DEF01 evidence and OBS02 limitation; confirm expected/actual tables and English wording. Discussion: [Actual questions, responses, decisions, corrections and responsible member]. Review outcome: [Actual outcome after corrections are checked].

11.2 Individual contributions

Member A: [Describe actual personal work, test IDs, evidence produced, corrections made and understanding of equivalence partitioning and boundary analysis].

Member B: [Describe actual personal work, independent checking, test IDs, evidence produced and lessons from testing].

11.3 Submission status

The English technical report and execution artifacts are prepared. Actual names, task allocation, meeting record and individual contributions are pending. No FTP submission or group approval is claimed. These administrative records must be completed before final coursework submission.

12 Source materials

[1] Course handout. Experiment 5 Using Black Box Testing in System Testing [Z]. Supplied original DOC, read on 22 October 2026.

[2] Project working documents. requirements.md, examples.md, E2 module structure, persistence-v0.4.md and csv-v0.5.md [R]. Local requirement and interface records.

[3] Project execution evidence. blackbox-e5.json, browser verification records and CSV regression record [CP]. Execution times and source hashes recorded in this report and its evidence directory.